Gavin Tranquilino

Mechatronics Engineering Student

gavintranquilino.com
gtranqui@uwaterloo.ca
github.com/gavintranquilino
linkedin.com/in/gavintranquilino

Education

University of Waterloo — **June 2028**
Candidate for BASc in Mechatronics Engineering — *Waterloo, ON*

- **Coursework**: (DSA) Data Structures and Algorithms, Linear Algebra, OOP, FPGA, PLC, Materials, Statistics, Ordinary Differential Equations, RTOS STM32, Microprocessors, Actuators and Power Electronics, Design and Dynamics of Machines

Experience

Mechatronics Engineering Assistant — January 2026 – April 2026
Acceleration Consortium — *Toronto, ON*

- Led integration of a self-driving lab, automating a 2D gantry and 6DoF robot arm to run material discovery experiments.
- Deployed an **MQTT** control broker and bridged wired/wireless links for reliable realtime **IoT** messaging.
- Architected the networking stack to autonomously orchestrate 8 **IoT devices**, enabling system control over **ethernet switching**.
- Used **Python** to develop **inverse kinematics** for the robot arm to transport materials autonomously.
- CAD modeled parts in **Fusion360** for manufacturing and prototyping housings for electronics, and tooling fixtures.

Humanoid Robotics Engineering Co-op — May 2025 – December 2025
WATonomous — *Waterloo, ON*

- Building **robotic arms** (6DoF) with tendon driven anthropomorphic hands (20DoF each), aiming for VR teleoperation.
- Containerized ROS2 system in **Docker**, mounting CAN transceivers to enable communication between subsystems.
- Designed **URDF** models to define the transform **TF tree** for **RL simulation** and training in **NVIDIA Isaac Sim**.
- Built visualization infrastructure connecting **Gazebo** simulations to **Foxglove** for real-time debugging and data analysis.
- Assembled **PCBs** with 0.5mm pitch **SMD components**, soldered by hand, reducing assembly costs by 30%

Undergraduate Research Assistant — September 2024 – December 2024
University of Waterloo Engineering IDEAs Clinic — *Waterloo, ON*

- Designed a digital CAD twin of an existing knee crutch in **SolidWorks**.
- Instrumented a wearable knee crutch, allowing force readings for gait analysis and material selection via **FEA**.
- Developed a **data acquisition** system using **I2C** and C++, converting a bathroom scale for real-time load measurements.
- Built **Python** scripts for force distribution visualization in **Matplotlib**, with data logging for **gait analysis**.

Mechanical Engineering Associate — January 2024 – April 2024
Sheartak Tools Ltd. — *Waterloo, ON*

- Designed 15 third party woodworking machinery upgrades with **DFMA** in **SolidWorks** to meet OEM specifications.
- Applied **GD&T** principles to guarantee manufacturing accuracy for custom machine parts.
- Created 25 detailed installation manuals, including parts lists and assembly instructions, ensuring ease of use for customers.

Projects

Autonomous LiDAR Navigation for Mobile Robot

- Developed **C++ ROS2** nodes to convert **LiDAR** data into a **2D costmap** for obstacle detection and perception.
- Generated a **world model** from costmap and odometry data to represent the current environment.
- Implemented **A* algorithm** to compute obstacle-aware paths through the mapped environment.
- Applied **Pure Pursuit** to follow planned paths for smooth differential drive navigation.

Warehouse Autonomous Guided Vehicles (AGV)

- **Won TMMC Software Challenge** by developing autonomous warehouse robots using **TurtleBot 4** and **ROS2**.
- Generated a real-time costmap converting 2D **LiDAR** scans to **occupancy grids** with obstacle inflation for perception.
- Implemented **CV** stop sign detection using **YOLOv8** with bounding box distance estimation to stop at intersections.
- Designed **cascading PID controller** for wall-following and heading control with **IMU** feedback for warehouse traversal.
- Solved collision risks by implementing **LiDAR safety zones** with emergency stopping and backward movement protocols.

Technical Skills

Software/Languages: Python, C, C++, CMake, SSH, Bash, Gazebo, Foxglove, Linux, Ubuntu, JS, HTML, CSS, SQL, LaTeX
Libraries/Frameworks: ROS2, Docker, OpenCV, MQTT, Git, MediaPipe, Flask, NumPy, OpenGL
Mechanical: SolidWorks, Fusion360, AutoCAD, GD&T, CAD, FEA, DFMA, 3D Printing, Machine Tools, Onshape
Electrical: I2C, SPI, UART, CAN Bus, RS485, RS282, Arduino, Raspberry Pi, Soldering, Oscilloscope, LiDAR, PLC, HMI